

Modelos de Deep Learning

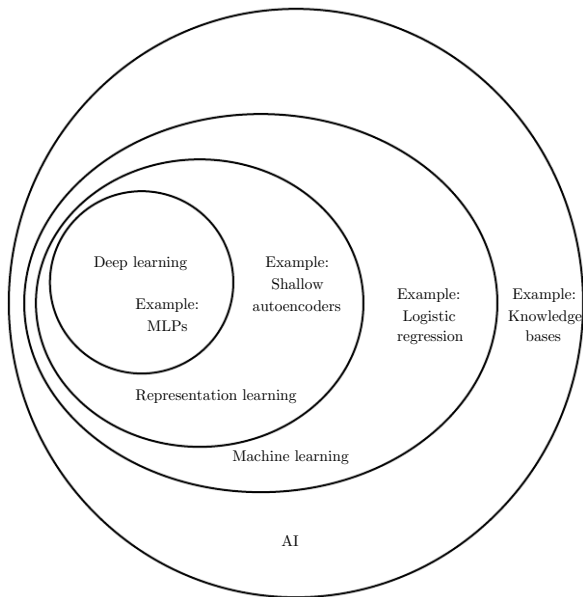
Introducción

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17 de agosto de 2024

Inteligencia Artificial (AI)



Machine learning - Algunos conceptos

Representación

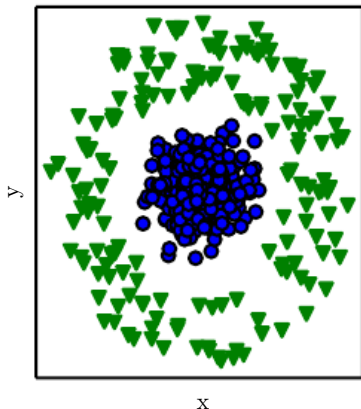
- ▶ Datos
- ▶ Hipótesis

Inferencia/generalización

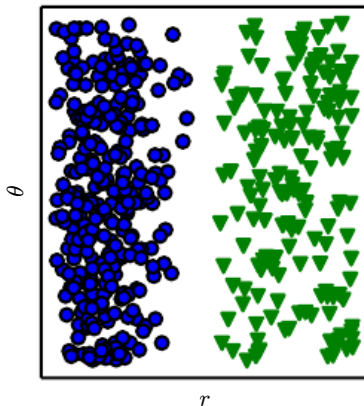
Representación

Punto $\longrightarrow$ sistemas de coordenadas

Cartesian coordinates



Polar coordinates



Uso: Visualización, Support Vector Machines (SVM) *kernels*

Representación

Frase $\longrightarrow$ ocurrencias de cada palabra

x_1 : The quick brown fox jumped over the lazy dog.

x_2 : The dog hunts a fox.

	the	quick	brown	fox	jumped	over	lazy	dog	hunts	a
x_1	2	1	1	1	1	1	1	1	0	0
x_2	1	0	0	1	0	0	0	1	1	1

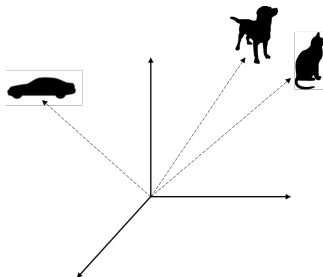
Uso: clasificación de documentos, e.g., detección de *spam*

Representación

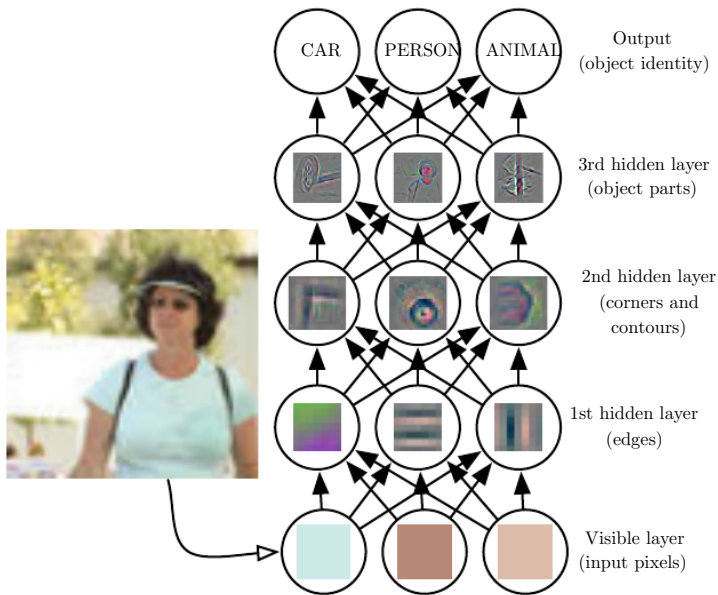
Palabra $\longrightarrow$ ocurrencias en cada documento

Document ID	1	2	3	4	5	6	7	8	9	10	11	12
Occurrences of 'dog'	1	0	2	0	4	6	3	0	6	2	0	1

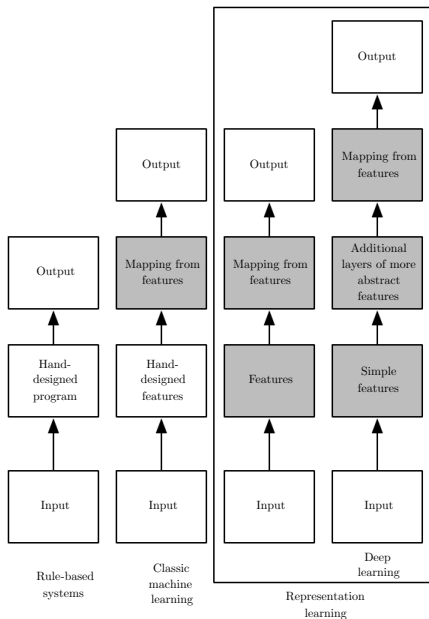
Uso: relación entre palabras por *similaridad* de vectores



Representación composicional

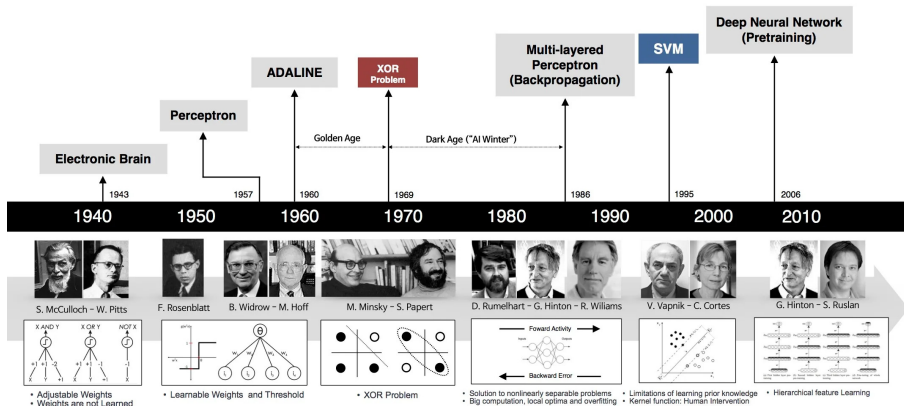


Disciplinas de AI



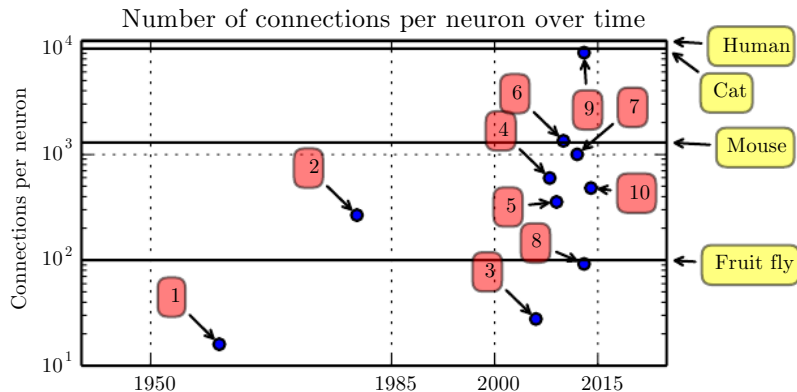
Deep Learning: Artificial Neural Networks

Hitos



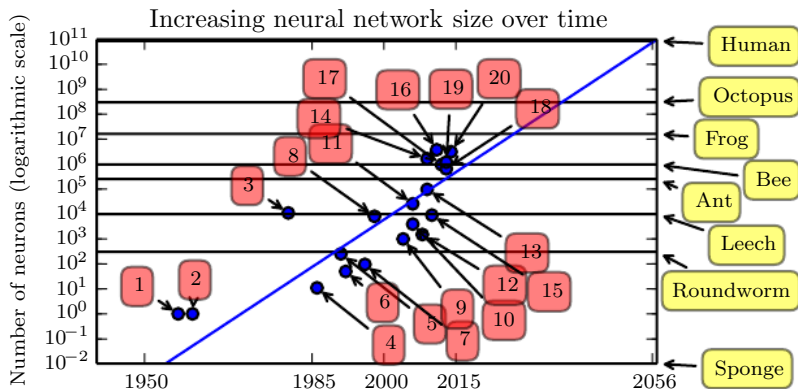
Artificial Neural Networks

Cantidad de conexiones



Artificial Neural Networks

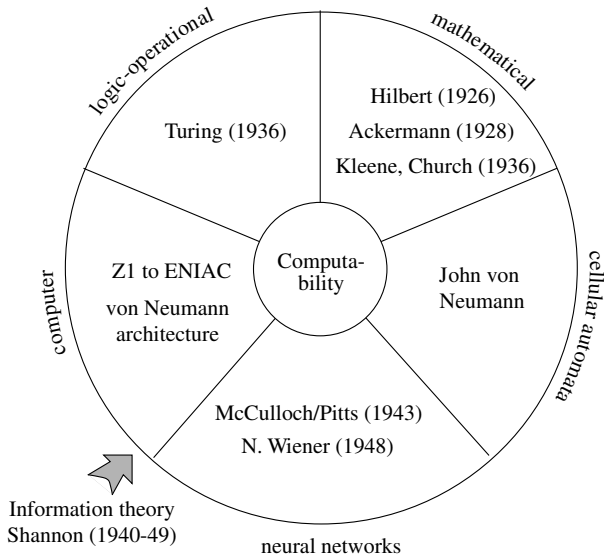
Cantidad de "neuronas"



VGG19 (2015): $1,4510^8$
GPT-3 (2020): $1,7510^{11}$

Deep Learning: Artificial Neural Networks

Modelos computacionales



Bibliografía

Deep Learning. Ian Goodfellow Yoshua Bengio Aaron Courville. MIT Press, 2016. Chapter 1.